Linear Delivery

Following the decision to embark upon a business change a key consideration is deciding on the delivery approach, often referred to as a system (or sometimes software) development lifecycle (SDLC). There are 2 basic types of SDLC, Linear and incremental, each of which have a subset of varying features.

Linear delivery lifecycles have distinct stages of project activity which are carried out in sequence, leading up to the full release of the new system using either parallel run or big bang implementation (possibly with pilot deployments). Either way, all of the software produced during the project is released at once and deployed along with new processes and staff support and training as required. The project is planned in advance and milestones are set out for the completion of each stage as well as the final deployment and post implementation review. Waterfall and the V model are 2 variances of the linear approach.

Waterfall

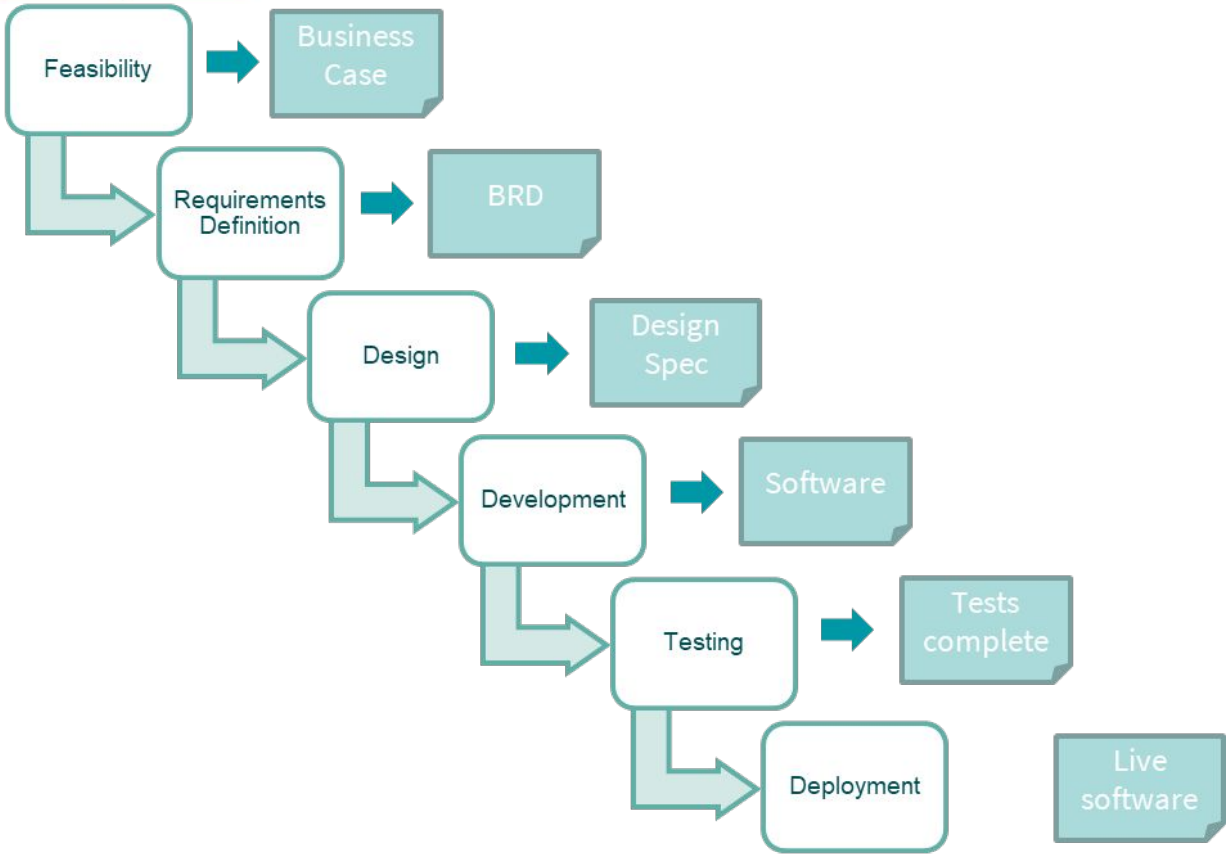
When IT change began to boom in the latter quarter of the 20th century, due to advancements in technology and cheaper computer memory. Many organisations were keen to gain greater efficiencies by introducing the latest tech, but IT system knowledge was severely lacking. Therefore, IT development teams took a lead on IT change. In many cases this resulted in the implementation of systems which adopted advancements in technology, but did not necessarily consider all of the business needs. Many systems were not completely fit for purposes, leaving organisations with many manual processes and workarounds, as well as a multitude of system enhancements if budgets were still available.

From here, many organisations aimed to shift the emphasis to IT enabled change, rather than IT driven change. Focus was given to strategic plans and studies of the business to understand needs more fully before investing in new technologies. Fearful of repeating previous mistakes, robust approaches were taken and project controls and governance were added to the change processes over time.

Waterfall

The waterfall approach was successful in addressing these issues and providing rigour that was needed. The main principle of the approach is that each stage is fully completed and signed off, before moving on to the next. The feasibility stage encompasses strategic decision making, investigation of the current situation and agreement on the objectives, scope and budget for the new business system (all POPIT areas considered). The business case is produced as part of this stage and remains live throughout. Project support structures were usually in place and a various roles such as business analyst and project manager emerged and evolved. Standard templates and development of best practice accompanied the approach until the required rigour was attained. Waterfall lifecycles worked well and, the highly structured approach allowed for easier project management and budgetary control, but there were drawbacks.

The Waterfall SDLC



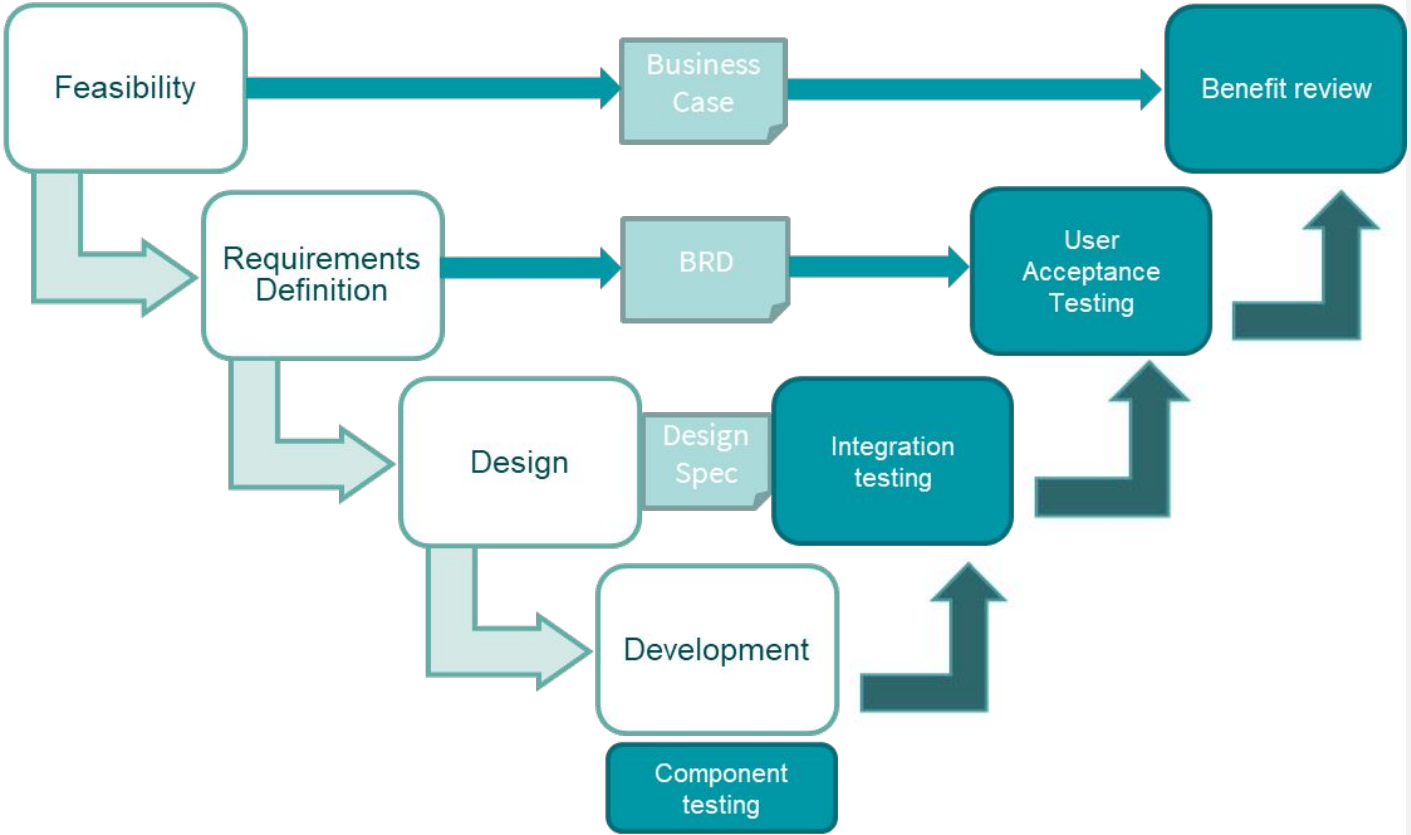
The Waterfall SDLC

One major drawback is that if mistakes were made in the earlier stages, such as missing or incorrect requirements, these mistakes were compounded as the incorrect requirements led to the design and development of features that wouldn’t meet business need. These errors may not be uncovered until the later stages of testing when business users carried out acceptance testing. These errors were costly to fix and revisiting them at such late stages has a large impact on budgets and delivery timescales. In some cases, changes to requirements could not be made because there was no budget, or time. Decisions as to whether to allow such changes were dealt with via strictly governed change control processes.

V Model

One answer to this problem was to make the requirements definition stage more robust, hence the term requirements engineering. Another was the introduction of the V-model, which uses the basic waterfall structure, but encourages early testing. The code cannot be tested until it is written, but test cases can be written beforehand and static reviews by testers of the outputs of each stage are compulsory when using this model, particularly focusing on the acceptance criteria.

V Model


Feasibility
Business Case
Benefit review
Requirements Definition
BRD
User Acceptance Testing
Design
Design Spec
Integration testing
Development
Component testing

V Model

The V-model, adopted by some organisations (but never as popular as the traditional waterfall model), helped to address the issues faced and both waterfall and v-model approaches were successfully used for the remainder of the 20th and into the 21st century and are still in use today. These approaches led to more steadfast change that met the business needs, but there are still major drawbacks. The primary issue is that they are slow! When in a fast-paced environment in which customer expectations, competitor actions and external events may lead to a desire to deliver value quickly and a need to adapt to a new situation, the waterfall approach is too slow. Organisations that which to respond quickly to change, simply could not adapt at the required pace.

VUCA

The VUCA model below shows the issues that organisations face, and that the waterfall approach simply wasn't designed to deal with.

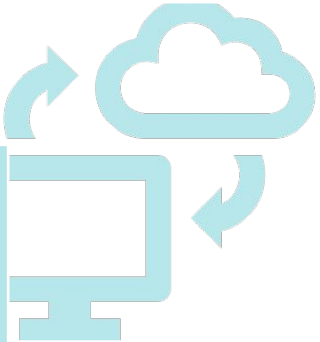
An answer to this issue has been to adopt an iterative approach to development and an incremental approach to delivery, which are features of Agile approaches.

Volatility

- The nature and speed of change

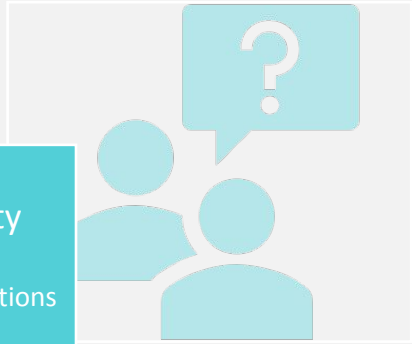
Uncertainty

- The lack of predictability

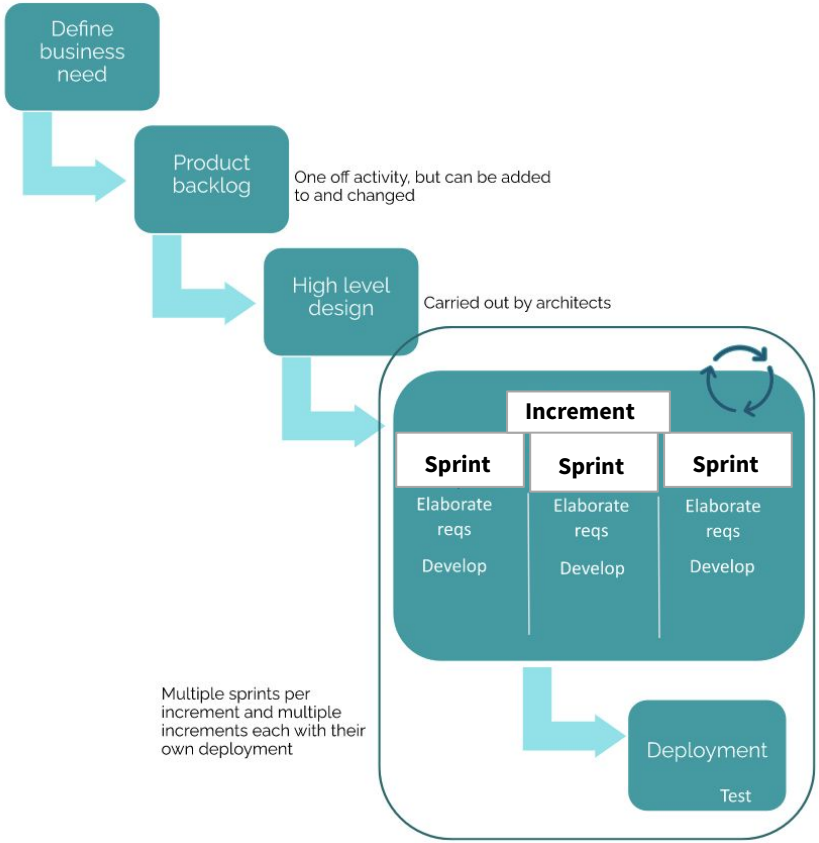


Complexity

- Multiple inputs, outputs and interfaces



Ambiguity

- Different interpretations

Iterative and Incremental Delivery